
Product Description / Equipment Specifications

Vantage® M284 Reverse Osmosis Units

General Description:

Vantage® M284 Series reverse osmosis units are designed and manufactured for industrial applications. They are horizontal configurations utilizing water cooled vertical pumps. They are pre-engineered, pre-assembled units that minimize installation and start-up costs. They are fully tested at the factory and require simple utility connections with minor set up and adjustment, and is ready for immediate on-line service.

Each unit is a two pass RO system mounted on an industrial coated structural steel frame. The RO unit includes a cartridge pre-filter, high pressure feed pump, FRP side entry pressure vessels, spiral wound thin film composite RO membranes, PVC low pressure feed and reject piping, stainless steel high pressure piping, and PVC product piping. The RO systems are controlled by a programmable logic controller (PLC) with a touch screen Human-Machine Interface (HMI). The Deluxe systems also utilize a Variable Frequency Drive (VFD) to control the RO feed pumps.

Mechanical Description:

The components of the *Vantage® M284 Series* reverse osmosis units are supported by an industrial coated carbon steel frame and designed in such a way to provide easy access for servicing, maintenance, and monitoring system performance. The systems are IBC 2006 (ASCE 7-05) seismic rated. Medium Pressure Thin Film Composite (TFC) membranes are provided as standard with an option for Low Pressure membranes for certain low temperature applications.

The RO feed, low-pressure reject, and product piping are schedule 80 PVC. The inlet and outlet piping connections to the unit are class 150 PVC flanges. The high-pressure piping is constructed of welded Sch10 316L stainless steel.

A single 316 stainless steel ball valve is used for pump discharge pressure control on the Plus RO units. The Deluxe RO unit utilizes a VFD to maintain a constant product flow. A 316L stainless steel globe valve is used to control the reject flow rate. Sample valves are provided for the RO feed upstream and downstream of the cartridge filter, membrane feed, reject, and combined product water. A sample valve is also located on the product port of each pressure vessel for analyzing individual vessel performance. Grooved/Flanged cleaning connections are provided at each stage of the RO to clean the membranes.

If an RO cleaning is required, a separate Evoqua Water Technologies CIP skid will be required to perform the on-site membrane cleaning or the membranes can be shipped to an approved Evoqua Water Technologies membrane cleaning facility for off-site cleaning.

Electrical Description:

A highly reliable control package is used to control the operational sequences of the *Vantage® M284 Series* reverse osmosis unit. The units are controlled by a PLC and operated via a color touch panel display (HMI). The Plus model has a motor starter and PLC housed in a single NEMA 4 enclosure. The Deluxe model has an integrally mounted VFD on the pump motor and PLC housed in a NEMA 4 enclosure. All systems have Ethernet connectivity for remote control and monitoring purposes. All systems require only a single power drop for operation. These controls monitor and protect the system from damage to itself, nearby equipment, systems, and/or personnel.

Operational Description:

Vantage® M284 Series reverse osmosis units operate continuously and are monitored by the instruments and a PLC. The PLC will alarm on non-critical conditions such as poor quality or shut down the RO system on critical conditions such as low suction pressure or high discharge pressure. All alarm and shut down conditions are indicated on the control interface. Dry contact inputs are provided for chemical tank low level, pretreatment interlock, clean water flush pump, storage tank high level, plus pump suction and pump discharge pressure switches. The pretreatment interlock or tank high input will put the RO into “Standby” mode. During the standby mode, the RO will periodically flush with RO permeate water to maintain optimal system performance.

Product Offering Overview:

Model Number M284R	Nominal Product Flow Rate*
024	50 GPM
044	100 GPM

*Product flow rates are based on the following; 1st pass 14 GFD and 2nd pass 21GFD average flux rates, well water with SDI<3, TDS = 1000 mg/l, pH = 7, Temperature = 50°F. Higher or lower product flow rates may be appropriate for other feed waters. Warmer or colder water or use of different membrane elements may require a different pump. An Applications Specialist can re-rate the unit or select an alternative membrane for other feed water conditions.

Nominal Design Parameters:

Configuration	Multi-Staged, Double Pass
Inlet Pressure	30 psig minimum
Inlet Temperature	50°F
Flux	1 st pass 14 GFD, 2 nd pass 21 GFD (average overall Flux 17.5 GFD)
Feed Water Source	Well
Feed Water Fouling Index	Silt Density Index (SDI) < 3
System Recovery	75% minimum
Product Pressure	35 psig

Recommended Operating Limits*:

Inlet Pressure	30 - 60 psig
Inlet Temperature	35 – 90°F
Flux	1 st pass 14-17 GFD, 2 nd pass 18-24 GFD
Feed Water Source	Well or Softened with Silt Density Index (SDI) < 3
System Recovery	75%

*Recommended operating limits are established as a guideline. Performance when operating within these parameters varies based on feed water conditions and the selected membrane. Under certain conditions, operation outside of these limits may be possible. A specific RO projection must be run for each individual application to ensure proper performance.

Parameters Not to Exceed:

Inlet Temperature	35°F Minimum - 112°F Maximum
Turbidity	1 NTU Maximum
Free Chlorine and/or Chloramines	<0.1 PPM Maximum
Overall System Recovery	85% Maximum

General Specifications:

Frame	
Materials	Structural carbon steel
Finish	Industrial coated, smooth finish, Blue, Federal Standard 25052
Cartridge Filter Housing	
Materials	304L SS, Passivated
Filter Rating/Material	5 micron Nom. DOE, Polypropylene
Size	7Rx3H (024 unit), 12Rx3H (044 unit)
Pump	
Manufacturer	Grundfos CRN series on Plus, Grundfos CRNE (integral VFD) on Deluxe, grooved or flanged connections
Materials	316 SS for product contact components
Motor	TEFC
Membranes	
Manufacturer	Hydramatics®
1 st Pass	Medium Pressure (CPA5-34) or Low Pressure (ESPA2-34) 400 ft ² active membrane surface area
2 nd Pass	Low Pressure 440 ft ² active membrane surface area (ESPA2-MAX)
Materials	Thin Film Composite (TFC)
Type	Spiral wound
Pressure Vessels/Relief	
Manufacturer	Protec
Model	8" Diameter, PRO-8-450 Series
Materials	Fiberglass reinforced plastic (FRP)
Rating	450 PSIG ASME Code
Process Connections	Grooved side entry feed/concentrate
Pressure Relief Valve	ASME code stamped, carbon steel body with stainless steel trim (wetted materials) and Viton soft seat

Piping Systems	
Feed piping	PVC, Schedule 80
High pressure piping	316L SS, Schedule 10, Passivated
Product piping	PVC, Schedule 80
Cleaning connections	Flanged/Grooved - feed & high pressure
Automatic Process Valves	
Inlet valve	Actuated EPDM lined SS disc butterfly valve, fail-to-close
Permeate flush inlet valve	Actuated EPDM lined SS disc butterfly valve, fail-to-close
Auto flush valve	Actuated ball valve, 316 SS, fail-to-close
Manual Valves	
Pump throttling valve	316 SS ball valve, socket weld, manual
Reject valve	316 SS globe valve, manual
Fast flush valve	316 SS ball valve, automated
Sample valves	¼" PVC valve(low pressure), ¼" 316 SS plug valve(high pressure), One for each vessel and one common ¼" PVC valve(product)
Product Divert Kit	

Product isolation	Actuated EPDM lined SS disc butterfly valve, fail-to-close
Product to drain	Actuated EPDM lined SS disc butterfly valve w/travel stops, fail-to-open
Final and Inter-Pass Product relief valves	100 psig, PVC body with stainless steel trim (normally wetted materials) and Viton soft seat.
Pneumatic speed control	Nickel plated brass, 1/4" tube fittings
Interstage Caustic Injection (Deluxe only)	
Retention Vessel	(1) empty 8" Diameter, PRO-8-450 Series
Solution Tank	HDPE with cover, mounted off-skid 22" diameter, 31" high, 35 gallon capacity
Metering Pump	Grundfos DDA high precision diaphragm metering pump with digital dosing and automatic motor speed adjustment
4-Function Valve	Anti-Siphon, pressure relief, back pressure regulator and pressure release (priming)
pH Instrumentation	Hach model SC100 PID controller and pH combination convertible sensor, general purpose glass electrode
Valves	(1) Parker 1/4" stainless steel plug valve
Interstage Caustic Injection Residence Time	Approx. 22 seconds

Instrumentation Specifications:

Pressure	
Indicators	316 SS, 63mm dial, FDA approved glycerin filled, 1/4" NPT
Low feed switch	United Electric 10 series, 316 SS / Buna-N, 120VAC, 4-50 PSIG, 1/4" MNPT
High discharge Switch (2nd Pass)	United Electric 10 series, 316 SS / Buna-N, 120VAC, 30-600 PSIG, 1/4" MNPT
High Product Pressure	United Electric 10 series, 316 SS / Buna-N, 120VAC, 30-600 PSIG, 1/4" MNPT
Interstage Pressure Transmitter	Ashcroft K1 series pressure transmitter, 316 SS, 4- 20 mA, 0- 150 PSIG, 1/4" MNPT
Quality/Temperature	
Conductivity/Temp monitor	Signet 8900 Series Multi-Parameter
Conductivity/Temp sensors	Signet 2850 Series
Flow	
Flow monitor	Signet 8900 Series Multi-Parameter
Flow sensors	Signet 2537

Controls Specifications for PLC packages (All models):

Main control panel	Carbon steel enclosure, frame mounted, NEMA 4, ANSI 61 gray (Indoor Installation Only)
Programmable Logic Controller (PLC) Options	Siemens S7-1215C (TIA Portal program-based) with built-in Ethernet connectivity Allen-Bradley option (MicroLogix 1100 with built-in Ethernet connectivity)
HMI - Touch Screen Interface Options	Siemens TP700 Comfort Panel with Ethernet connectivity Allen-Bradley PanelView Component 600 color touch
Shutdown alarms	High product conductivity/low salt rejection % (final) Low feed pressure Low feed flow High product pressure (1 st Pass product and final) High pump discharge pressure (1 st pass, 2nd Pass) High feed water temperature Emergency Stop VFD fault (Deluxe)
HMI status indicator/value	All Alarms Feed flow, reject flows, product flow, % recovery, % salt rejection, final product conductivity Total run time RO operating mode (Flush, Rinse, Service, Standby) Pump status (% VFD on Deluxe models) Inlet, reject, product to tank, product to drain valve status Pretreatment lockout Storage tank full (Standby – no call for water)
HMI Switches / Pushbuttons	RO selector Man/Auto RO selector Start/Stop Auto flush selector On/Off Alarm silence Alarm reset
Miscellaneous controls	Alarm horn and alarm pilot light Chemical injection pump terminals Auxiliary contacts for pump running, clean water flush active, & fault
Solenoid Valves	Stacked MAC-181B 24VDC

Interface Communication Specifications:

Pretreatment system - out of service	Activation of this signal confirms that the pretreatment equipment is available to deliver water to the RO unit. (Dry contacts must be closed in order to operate)
Clean Water Flush	Activation of this signal by the PLC initiates product water pump activation (Dry contact closes during clean water flush)
Call-for-water signal	Activation of this signal confirms that the post treatment equipment (storage tank or other equipment) is requesting RO product water. (Dry contact closes for call for water)

Factory Procedures:

Assembly	Fully assembled at the factory
Wiring	Fully wired at the factory
Testing	Hydrostatic pressure test, Electrical integrity test, Factory

	simulated functional test
Membrane Elements	Membrane elements are shipped in their original packaging for installation on site at time of start-up
Shipping preparation	Heat shrink protection (some components may be removed for protection prior to shipment)

Regulations and Standards:

Facility Quality System	ISO 9001
Stainless Steel Pipe Finish	Interior - Mill Exterior – Bead Blast
Seismic Rating	IBC 2006 (ASCE 7-05) Occupancy Category II Site Class D Seismic Design Category D Sds = 1.0 Fp = 0.40 Wp
Frame Welding	AWS D1.1
Electrical and Controls	NRTL Marked (cETLus), National Electrical Code (NEC)
Panel NEMA rating	NEMA 4 (Indoor Installation Only)

Documentation Package:

Documents	Installation procedures, start-up procedures, operation procedures, controls write-up, specifications, spare parts list, and Technical Service and Support services.
Drawings	Process and Instrument diagram (P&ID), General Arrangement Drawing (GA), Electrical Schematics
Software	Installed at factory prior to testing
Quality Documents	Quality Assurance data reports

Nominal Flow Rate Specifications*:	Model Number	
Description	024	044
Nom. Product (GPM)	50	100
Nom. Feed (GPM)	66	128
Nom. Reject (GPM)	15	28

*Flows based upon 14 GFD.

Configuration Specifications:	Model Number	
Description	024	044
Vessel array staging	2:1:1//1:1	4:2:1//2:1:1
Membranes / vessel	4	4
Membrane quantity	24	44
Vessel quantity	6	11
Pump model (Plus)	CRN15-12	CRN32-8
Pump model (Deluxe)	CRNE15-12	CRNE32-8

Cleaning Connection Specifications:	Model Number	
Description	024	044
Feed (Grooved) <i>Plus</i>	2"	2"
Feed (Flanged) <i>Deluxe</i>	3"	3"
Interstage (Grooved)	2"	2"
Reject (Grooved)	2"	2"
Product (Flanged)	1.5"	1.5"

Customer Connection Specifications*:	Model Number	
Description	024	044
Feed	3"	3"
Product	2"	3"
Reject	1.5"	1.5"

* All connections are Class 150 ANSI flanges

Utility Requirements:	Model Number	
Description	024	044
High voltage service	460VAC	
High voltage FLA (460)	70 Amps	106 Amps (Plus) 82 Amps (Deluxe)
Motor HP (ref.)	25 x 2	40 x 2 (Plus) 30 x 2 (Deluxe)
Waste drain* - gpm	80	130
Clean, Oil-Free, Dry Air	2 SCFM @ 80-100 psig	

*Flows based upon 2nd pass average 21 GFD. If flux is higher than 21 GFD, increase flows accordingly. Install a floor drain (in addition to a waste drain) for general maintenance purposes.

Physical Dimension Specifications:	Model Number	
Description	024	044
Length (in)	226	226
Width (in)	54	54
Height (in)	90	90
Est. Ship weight (lbs) (Heaviest Option Package)	5576	6115

Standard Product Ordering Information:

		M284 R	02 4	D	M	S	D
M284R		VANTAGE M280 SERIES RO					
DESIGN FLOW							
024		24 MEMBRANES (4M, 50GPM)					
044		44 MEMBRANES (4M, 100GPM)					
TRIM PACKAGE CHOICE							
P		PLUS					
D		DELUXE (with interstage Caustic Injection System)					
MEMBRANE CHOICE							
M		Hydranautics CPA5-34) (Second Pass ESPA2-MAX)					
L		Hydranautics ESPA2-34 (SECOND PASS ESPA2-MAX)					
CONTROL PKGS							
S		SIEMENS PLC/HMI 460 VAC / 3 PH / 60 HZ					
A		ALLEN-BRADLEY PLC/HMI 460 VAC / 3 PH / 60 HZ					
VOLTAGE OPTION							
D		460 VOLT, 3 PHASE, 60HZ					

Note: For non-standard options, contact Evoqua Water Technologies Applications Engineering.